

87%

2nd year End of Year test

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Date: 12/5/26

Question 1.

- (a) Work out the value of $3p - 4t^2$, when $p = 6$ and $t = 5$.

$$\begin{array}{l} 3p - 4t^2 \\ 3(6) - 4(5)^2 \\ 18 - 100 \\ 18 - 100 \\ -382 \end{array} \quad \begin{array}{l} 18 - 4(25) \\ 18 - 100 \\ -82 \end{array} \quad \frac{3}{5}$$

- (b) Multiply out and simplify $(2x - 3)(4 - 5x + x^2)$.

$$\begin{array}{l} 6x - 10x^2 + 2x^3 - 12 + 15x - 3x^2 \\ 21x - 13x^2 + 2x^3 - 12 \end{array} \quad \checkmark 18$$

- (c) Factorise fully $5fh - 2h^2 - 6h + 15f$.

$$\begin{array}{l} 5f(h+3) - 2h(h+3) \\ (5f-2h)(h+3) \end{array}$$

$\checkmark 18$

Question 2.

- (a) Solve the equation $5x - 10 = 3x + 2$.

$$\begin{aligned} 2x - 10 &= 2 \\ 2x &= 12 \\ x &= 6 \end{aligned}$$

✓ 15

- (b)

Solve the simultaneous equations:

$$2x - 3y = 18$$

$$5x + 9y = -10$$

$$\begin{aligned} 2x - 3y &= 18 \quad (5) \\ 5x + 9y &= -10 \quad (-2) \end{aligned}$$

$$10x - 15y = 90$$

$$-10x - 18y = 20$$

$$-33y = 110$$

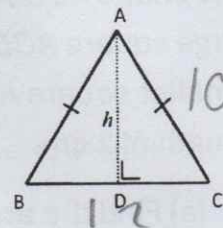
$$33y = -110$$

$$y = -\frac{10}{3}$$

✓ 7

Question 3.

The isosceles triangle ABC is shown in the diagram. The base $|BC| = 12 \text{ cm}$ and the side $|AC| = 10 \text{ cm}$.



- (a) Find h , the perpendicular height of the triangle.

$$\begin{aligned} 6^2 + h^2 &= 10^2 \\ 36 + h^2 &= 100 \\ h^2 &= 64 \\ h &= 8 \end{aligned}$$

✓ 10

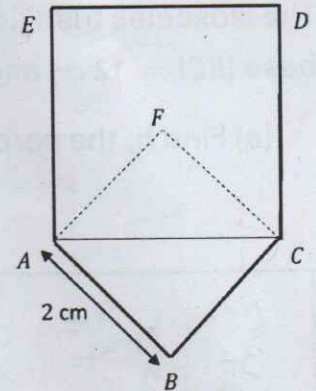
- (b) Explain why the triangle ABD and triangle ACD are congruent. Give a reason for each statement that you make.

$$\begin{aligned} |AC| &= |AB| \dots \text{given} \\ |BD| &= |DC| \dots \text{each are half of BC, we know this because } \angle ADC \\ &\text{is given as a right angle} \\ |AD| &= \text{common side} \dots \text{both triangles share AD} \\ \therefore &\text{congruent by SSS} \end{aligned}$$

✓ 10

Question 4.

The shape $ABCDE$, shown in the diagram, consists of a large square $ACDE$ standing on the diagonal $[AC]$ of a smaller square $ABCF$. The smaller square has a side length of 2 cm.



(a) Find the **area** of the shape $ABCDE$.

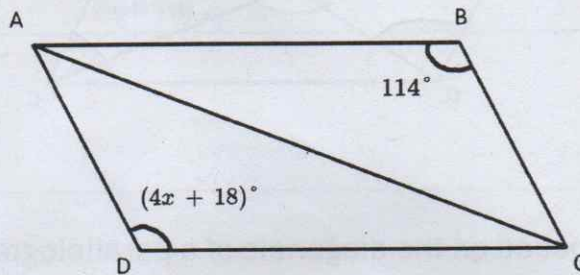
$$\begin{aligned}
 2^2 + 2^2 &= |AC|^2 \\
 4 + 4 &= |AC|^2 \\
 8 &= |AC|^2 \\
 |AC| &= 2\sqrt{2} \\
 2\sqrt{2} \times 2\sqrt{2} &= 8 \text{ cm}^2 \\
 2 \times 2 &= 4 \\
 4 \div 2 &= 2 \text{ cm}^2 \\
 8 + 2 &= 10 \text{ cm}^2 \\
 \text{Answer} &= 10 \text{ cm}^2
 \end{aligned}$$

(b) Find the **perimeter** of the shape $ABCDE$. Give your answer corrected to 2 decimal places.

$$\begin{aligned}
 2\sqrt{2} + 2\sqrt{2} + 2\sqrt{2} + 2 + 2 \\
 12.49 \text{ cm}
 \end{aligned}$$

Question 5.

The figure ABCD is a parallelogram. In the diagram, $|\angle ABC| = 114^\circ$ and $|\angle ADC| = (4x + 18)^\circ$.



$$\angle B = \angle D$$

- (a) Find the value of x based on the angles of the diagram. Justify your answer with a theorem or axiom. *vertically opposite? alternate?*

$$4x + 18 = 114 \quad \checkmark \text{ alternate angles}$$

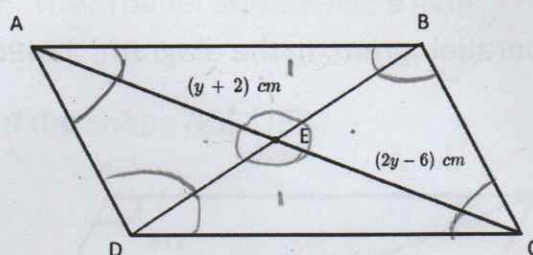
$$4x = 114 - 18$$

$$4x = 96$$

$$x = 24$$

3

- (b) Two lines are drawn diagonally in the parallelogram and intersect at point E where $|AE| = (y + 2) \text{ cm}$ and $|EC| = (2y - 6) \text{ cm}$.



all of the angles work out, therefore they must be equal, right?

Find the value of y based on the diagonals of a parallelogram.

for 1946
6.2

$$\begin{aligned}
 y + 2 &= 2y - 6 \\
 y &= 2y - 8 \\
 -y &= -8 \\
 y &= 8
 \end{aligned}
 \qquad
 \begin{aligned}
 y + 2 &= 2y - 6 \\
 y + 8 &= 2y \\
 8 &= y
 \end{aligned}$$

10

$$\begin{aligned}
 y + 2 &= |AE| \quad (2) \\
 2y - 6 &= |EC| \quad (-1) \\
 y &= 8 \\
 2y - 6 &= 2(8) - 6 \\
 2y - 6 &= 16 - 6 \\
 2y - 6 &= 10 \\
 2y &= 10 + 6 \\
 2y &= 16 \\
 y &= 8
 \end{aligned}$$

$$y + 2 = 2y - 6$$

$$2 = y - 6$$

$$-4 = y$$

$$y = -4$$

Something is wrong here, what am I doing wrong?